TABLE 13. Model performance in the SCRAIM dataset depending on the usage of the task description.

Target variable With(out) task description	Max. R^2	Min. MAE	Min. RMSE	Min. MMRE
Effort				
With task description	**0.51**[1]	**2.61**[1]	**5.29**[1]	**0.64**[2]
Without task description	0.35	2.77	5.93	0.85
Duration				
With task description	**0.37**[1]	**1.88**[2]	**3.23**[1]	**0.56**[3]
Without task description	0.00	2.31	4.05	0.83

1 - Random Forest; 2 - XGBoost; 3 - Support Vector Regression

Instead, two estimation models were built for each target variable and algorithm, one including the task description in the input variables (tokenized) and the other excluding the task description, and their predictive performance was compared.

The performance results are summarized in Table 13. For each metric and experiment, it is shown the value of the best-performing algorithm. The detailed results for the different algorithms, using the task description, have been presented before in Tables 3 and 4.

The results obtained using the task description in the training features presented much better results. In fact, this feature is so important that without it all the R^2 scores were negative or 0 in the duration estimation model, which means that, without the task description, no algorithm can explain the relationship between the input variables and the task duration. The elimination of other features from the set of input variables led to smaller performance degradation, so it is possible to conclude that for both SCRAIM models (task duration and task effort estimation), the task description feature is the most important one.

E. COMPARISON WITH RELATED WORK AND EFFECT OF ESTIMATION GRANULARITY

As pointed out in section II, most of the existing studies seek to make estimates at the project level, not at the level of individual tasks or issues, as we do in this article, so the performance results are not directly comparable. Besides that, most of the studies take advantage of size or complexity estimates as input for project effort and duration estimation, whilst we don't assume the existence of any size or complexity estimates (e.g., story points) to help predict the effort or duration required by a task or issue.

For example, Pospieszny et al. [5] achieved very good MMRE scores (0.13-0.15) and PRED(30) scores (81-82%) in predicting overall project effort and duration in an ISBSG dataset with 1,192 projects. Their input variables included the project's relative size in function points grouped into categories (similar to T-shirt sizes) – usually an important predictor of the project effort and duration. An additional difference to our work is that they log-transformed the target variables, but did not apply the reverse transformation when computing the performance metrics, which possibly led to overly optimistic scores, as acknowledged by the authors.

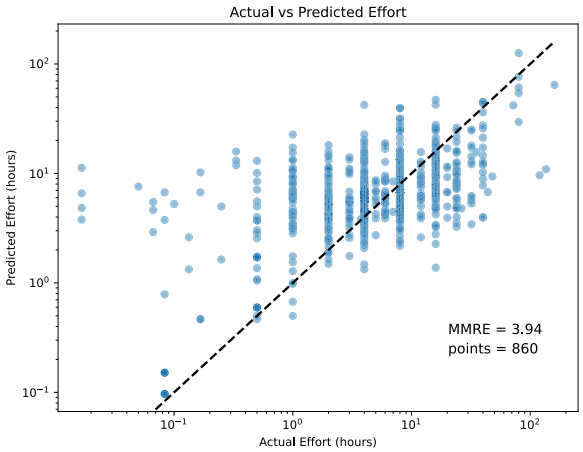


FIGURE 2. Actual versus predicted effort by issue in the JIRA dataset.

As mentioned in section II, there are some studies that try to predict the resolution time of newly reported issues or bugs. Ardimento and Mele [21] try to predict if the resolution of a bug will be 'slow' or 'fast', but not the actual resolution time (or duration) as in our case. Al-Zubaidi et al. [22] try to predict the resolution time of newly reported issues, representing feature requests, bug reports, etc. but don't report metrics suitable for comparison across different datasets.

Because of the error compensation that occurs when individual estimates are aggregated (*e.g.*, when summing the effort estimates of individual tasks to arrive at an effort estimate for the whole project), it is logical that the MRE of the aggregated project estimate is smaller than the MRE of individual task estimates, especially if the individual estimates are unbiased and independent.

In general, obtaining accurate effort and duration estimates at the level of individual tasks or issues is much more challenging, and very few works address such challenges. In practice, individual effort estimates with a small accuracy but a good balance between over and under-estimates may lead to accurate estimates at the project level - the most important ones for making commitments with customers. This phenomenon can be observed even in our dataset that presented inferior effort estimation performance - the JIRA dataset, as depicted in Figures 2 and 3.

Figure 2 illustrates the correlation between actual effort values and those predicted by a Random Forest Regressor in a test dataset. The actual values display a certain degree of discretization as time logging is often performed in multiples of hours. In contrast, each point in Figure 3 represents a unique project within the test dataset; the 'x' and 'y' coordinates indicate the cumulative actual and predicted effort values for the specific project, respectively. The differences in estimation accuracy are striking, either by visual inspection or by observing that the MMRE drops from 3.94 to 0.23 when task effort estimates are aggregated to the project level.

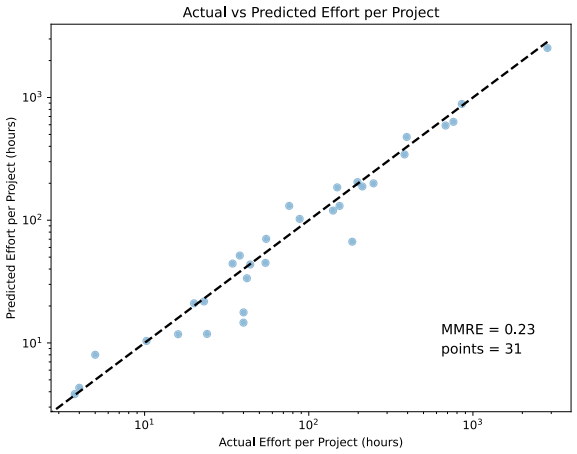


FIGURE 3. Actual versus predicted effort by project in the JIRA dataset.

TABLE 14. Top-performing algorithm for each dataset and target variable.

Data set	Target variable	Best algorithm	Normalized score
SCRAIM	Duration	XGBoost	92.12
SCRAIM	Effort	Random Forest	98.88
JIRA	Duration	XGBoost	98.00
JIRA	Effort	Random Forest	85.96
Project Control	Duration	Extra Trees Regressor	99.86
Project Control	Effort	Extra Trees Regressor	99.45
overall	overall	XGBoost	92.02

F. ANSWERS TO RESEARCH QUESTIONS

We next try to answer our initial research questions.

RQ1: What are the best-performing ML algorithms for task effort and duration estimation?

Table 14 shows the top-performing algorithm for each dataset and target variable, based on normalized scores computed as follows: (i) for each experiment and metric, the best-performing algorithm is given a normalized score of 100; (ii) the remaining algorithms are given normalized scores relative to this best-performer; (iii) finally, each algorithm's overall score is calculated by averaging its normalized scores across all four metrics we're considering.

The top-performing algorithms are all ensemble-based – Random Forest, Extra Trees Regressor and XGBoost. Overall, across all datasets and target variables, XGBoost is the best-performing, with an average normalized score of 92,02, closely followed by the other ensemble-based algorithms (with values between 88.70 and 92.02), and at a significant distance to the best-performing non-ensemble algorithm (Supper Vector Regression, with 72.19).

Our findings across three datasets suggest that ensemble-based algorithms, like Random Forest, Extra Trees Regressor and XGBoost, are the best choice for task effort and duration estimation.

RQ2: How effective are ML techniques in predicting the effort and duration of individual tasks or issues of software projects in production environments?

TABLE 15. Effectiveness of the top-performing algorithm for each dataset and target variable.

Dataset	Target v.	#Samples	#Inp.v.	R^2	MMRE
SCRAIM	Duration	853	6	0.32	0.64
SCRAIM	Effort	853	6	0.51	0.67
JIRA	Duration	4,094	9	0.37	10.27
JIRA	Effort	3,435	9	0.36	5.62
Project Control	Duration	54,153	15	0.96	0.11
Project Control	Effort	54,153	16	0.90	0.36

Table 15 shows, for each dataset and target variable, the scores obtained by the top-performing algorithm (see Table 14) for the metrics that are comparable across datasets (R^2 and MMRE), as well as the number of samples in each dataset and the number of input variables used for training.

Overall, the coefficient of determination (R^2) and the estimation accuracy, as measured by the MMRE, varied significantly across the datasets, with the best results achieved with the Project Control dataset.

Taking as reference the thresholds proposed by Jorgeensen and Shepperd [20] for classifying an estimation model accurate (MMRE ≤ 0.25 and PRED(0.3) $\geq 75\%$), the estimation accuracy of our models can be classified as good in the third dataset, poor in the second dataset, and intermediate in the first dataset.

However, it is important to note that these thresholds have been primarily proposed and applied for estimates at the project level, as opposed to our focus on individual tasks or issues. In Section IV-E, we demonstrate that even when the accuracy of individual estimates is poor, the aggregated estimates at the project level may present a good accuracy due to error compensation.

Hence further research is needed to determine the best metrics and thresholds for assessing the accuracy of effort and duration estimation models at the level of individual tasks or issues in software projects.

Regarding the causes of performance variation across the different datasets, the results show that having more data samples doesn't always result in better models. Even though the models trained with the Project Control dataset (which had the most data samples and training features) performed the best, models trained with the SCRAIM dataset performed better than those trained with the JIRA dataset, despite having fewer data samples and training features.

Therefore, other factors, besides the amount of data or the number of features, also play a significant role. For instance, the JIRA dataset contains issues of very diverse granularity, with effort ranging from as little as 1 minute to as much as 240 hours. If the predicted effort for a 1-minute task is 1 hour, its MRE will be 60. Given that the MMRE is extremely sensitive to individual predictions with excessively large MRE, this dataset's resulting high MMRE of 5.62 isn't surprising. On the other hand, the SCRAIM dataset covers a narrower range of effort, from 30 minutes to 147 hours, potentially contributing to more accurate overall predictions and a smaller MMRE. These observations suggest that the degree